

Connor Beatty

PHYS305

Foucault's Pendulum: Effect of the Coriolis Force

Introduction

Foucault's pendulum is an old demonstration showing that the Earth rotates. A pendulum swinging in a rotating reference frame will feel some sideways acceleration due to the Coriolis force, which causes the direction of oscillation to change over time. In this project I studied the motion of the pendulum in the rotating frame using the coupled equations of motion for both $x(t)$ and $y(t)$ which include both Coriolis terms and centrifugal terms.

To solve the equations a little easier, I rewrote the equations in dimensionless form using the natural pendulum timescale $\tau = \sqrt{\ell/g}$ and the dimensionless time $\tilde{t} = t/\tau$. This gives the Rossby number $R_0 = \sqrt{g/\ell}/\Omega$ which compares the rotation timescale to the swing timescale. Then I converted the dimensionless second order system into a first order system and solved it numerically using the Runge-Kutta (RK4) method. Before just jumping in and trying to code everything, I checked the correctness of the solver by doing a self-convergence test for a certain R_0 and estimating some numerical errors using Richardson extrapolation. Finally, I compared the trajectories and time series for $R_0 = 10, 2$, and 1 to basically just see how the Rossby number changes how the pendulum swings.

Governing Equations

In a reference frame that's rotating with angular velocity Ω (just the Earth) the small angle motion of a pendulum bob can be written in terms of its displacements in the $x(t)$ and $y(t)$ as:

$$\ddot{x} = -\frac{g}{\ell}x + 2\Omega\dot{y} + \Omega^2x, \quad \ddot{y} = -\frac{g}{\ell}y - 2\Omega\dot{x} + \Omega^2y$$

The terms with the $-\frac{g}{\ell}$ are the normal restoring accelerations for a pendulum in the small angle limit. The terms that go with the velocities, $2\Omega\dot{y}$ and $-2\Omega\dot{x}$, come from the Coriolis force and coupled motions. The terms proportional to Ω^2 are the centrifugal parts. These extra rotation terms are what cause the oscillation path to change from a straight looking line and make a slowly changing trajectory in the x - y plane giving it different looking patterns.

Non-dimensionalization

To reduce the number of parameters and make the numerical integration easier I make the equations dimensionless using the natural pendulum timescale and dimensionless time:

$$\tau = \sqrt{\frac{l}{g}} \quad \tilde{t} = \frac{t}{\tau}$$

Then sing the chain rule, derivatives transform as:

$$\frac{d}{dt} = \frac{1}{\tau} \frac{d}{d\tilde{t}} \quad \text{and} \quad \frac{d^2}{dt^2} = \frac{1}{\tau^2} \frac{d^2}{d\tilde{t}^2}$$

So...

$$\dot{x} = \frac{1}{\tau} \frac{dx}{d\tilde{t}} \quad \text{and} \quad \ddot{x} = \frac{1}{\tau^2} \frac{d^2x}{d\tilde{t}^2}$$

Also, similarly for $y(t)$ substituting these into the dimensional equations and multiplying through τ^2 gives:

$$\frac{d^2x}{d\tilde{t}^2} = -x + 2\Omega\tau \frac{dy}{d\tilde{t}} + (\Omega\tau)^2 x \quad \text{and} \quad \frac{d^2y}{d\tilde{t}^2} = -y - 2\Omega\tau \frac{dx}{d\tilde{t}} + (\Omega\tau)^2 y$$

At this point the rotation comes only from the $\Omega\tau$ stuff. I can define the Rossby number by:

$$\frac{1}{R_0} = \Omega\tau = \Omega \sqrt{\frac{l}{g}} \quad \text{and} \quad R_0 = \frac{\sqrt{g/l}}{\Omega}$$

So the final dimensionless equations become:

$$\frac{d^2x}{d\tilde{t}^2} = -x + \frac{2}{R_0} \frac{dy}{d\tilde{t}} + \frac{1}{R_0^2} x \quad \text{and} \quad \frac{d^2y}{d\tilde{t}^2} = -y - \frac{2}{R_0} \frac{dx}{d\tilde{t}} + \frac{1}{R_0^2} y$$

First-order system and initial conditions

To integrate the equations using RK4, I rewrote the second-order system as a first-order system by introducing the variables:

$$u = x, \quad v = y, \quad p = \frac{dx}{d\tilde{t}}, \quad q = \frac{dy}{d\tilde{t}}$$

With these, the system becomes:

$$\frac{du}{d\tilde{t}} = p, \quad \frac{dv}{d\tilde{t}} = q$$

$$\frac{dp}{d\tilde{t}} = -u + \frac{2}{R_0} q + \frac{1}{R_0^2} u \quad \text{and} \quad \frac{dq}{d\tilde{t}} = -v - \frac{2}{R_0} p + \frac{1}{R_0^2} v$$

The initial conditions I was given for this project were:

$$x(0) = 1, \quad y(0) = 0, \quad \dot{x}(0) = 0, \quad \dot{y}(0) = 0$$

Which look like this in the first order:

$$u(0) = 1, \quad v(0) = 0, \quad p(0) = 0, \quad q(0) = 0$$

Using RK4 and Integration time

I solved the first-order system using the RK4 method in dimensionless time. The state vector $\mathbf{y} = (x, y, v_x, v_y)^T$ is followed by $\tilde{t}_n$ to $\tilde{t}_{n+1} = \tilde{t}_n + h$ and using four intermediate slope evaluations of $\mathbf{y}' = \mathbf{f}(\tilde{t}, \mathbf{y})$ giving fourth-order accuracy in the step size h . So, to choose the best integration interval I used the dimensionless time scales that we given in the project:

$$\widetilde{T_p} = 2\pi \quad \text{and} \quad \widetilde{T_\omega} = 2\pi R_0$$

For each R_0 , I integrated long enough to capture the longer timescale so the motion and resulting trajectory could be compared with $R_0 = 10, 2, 1$

Convergence and Richardson error

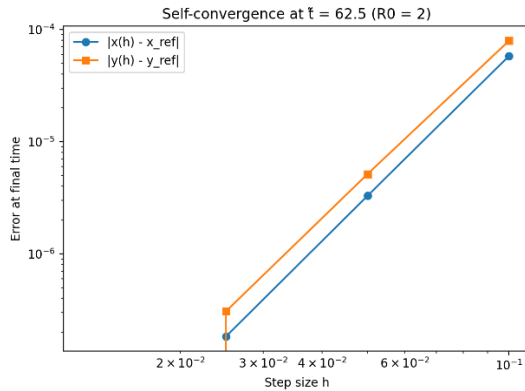
Before going through all the trajectories I checked that RK4 was giving good results by a self-convergence test at $R_0=2$. I integrated to $\tilde{t} = 62.5$ with step sized $h = 0.1, 0.05, 0.025$, and 0.0125 then compared the final values of $x(\tilde{t})$ and $y(\tilde{t})$ to the finest step result just for a good reference. On a log-log plot of error versus h (Figure 1) the errors decrease with a slope that goes with the 4th order behavior like we expect for RK4. To get a better quantitative error estimation I used Richardson extrapolation with order $p=4$, and by using the solutions computed with h and $h/2$ the extrapolated value is:

$$S_R = S_{h/2} + \frac{S_{h/2} - S_h}{2^p - 1}$$

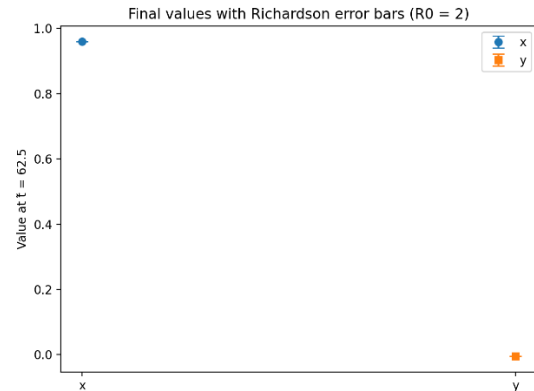
And the error estimate is $\Delta S \approx |S_R - S_{h/2}|$. Then at $\tilde{t} = 62.5$ for $R_0=2$ we get very small error estimates:

$$\Delta x \approx 1.23 \times 10^{-8}, \quad \Delta y \approx 2.05 \times 10^{-8}$$

These errors are a lot smaller than what the scale is on the plot which is why the error bars are basically invisible on the Richardson plot. (Figure 2). For the trajectory and time-series figures ($R_0 = 10, 2, 1$), I used $h = 0.01$ and integrated to $\tilde{t}_{\text{end}} = 20\pi \approx 62.83$. For the convergence test at $R_0 = 2$, I used $\tilde{t} = 62.5$ with $h = 0.1, 0.05, 0.025, 0.0125$.



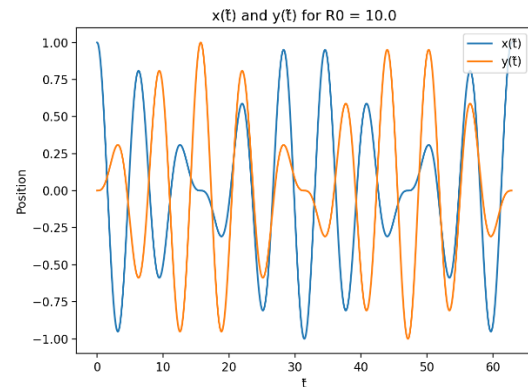
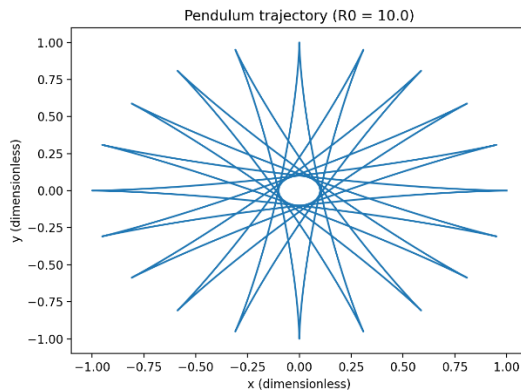
(Figure 1)



(Figure 2)

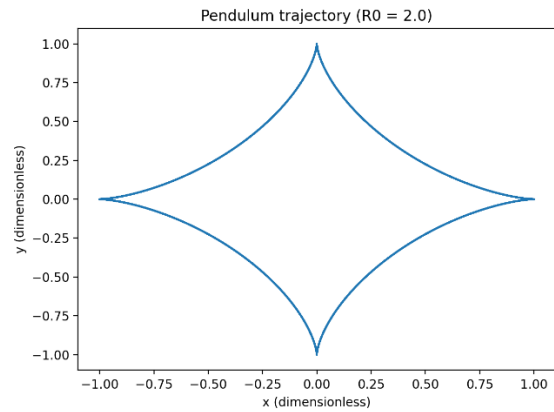
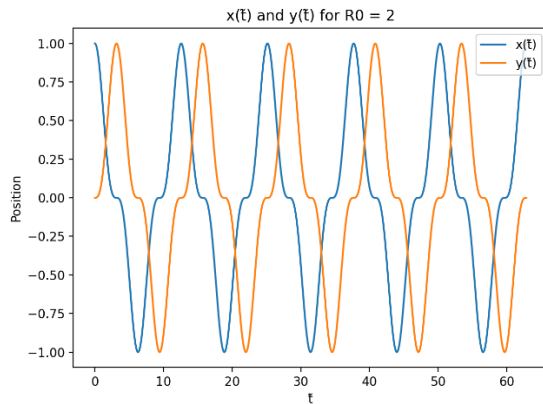
Results for different R0

R0=10



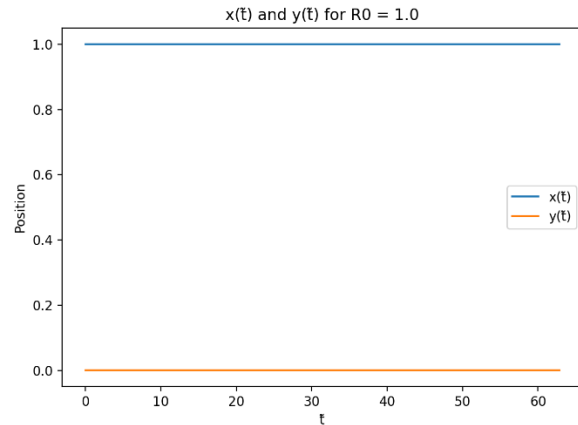
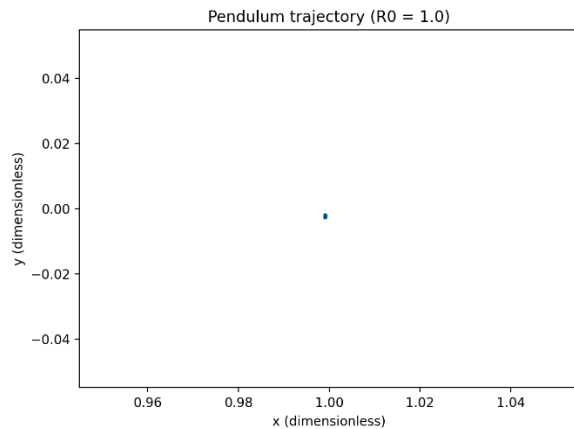
The path makes a star pattern where the motion looks like repeated, basically straight swings with direction slowly changing over time. This means that, even though the pendulum is doing roughly periodic oscillations, the plane of oscillation is rotating in the rotating-frame, so the direction of the back-and-forth motion drifts and goes over multiple angles.

R0=2



The motion looks very 2D, and the oscillations look less “simple sinusoid” than the large R_0 case. The trajectory forms diamond looking orbit. Compared to $R_0 = 10$, the path is more organized into a single repeating loop, which shows that the rotation effects are strong enough to shape the orbit into a pretty distinct 2D pattern. Overall, decreasing R_0 increases the importance of rotation for how the bob moves.

$R_0=1$



The x - y trajectory looks like single point meaning the pendulum does not swing for these initial conditions. This happens because at $R_0 = 1$ the restoring and centrifugal terms cancel in the dimensionless equations, and with zero initial velocity the Coriolis terms are also zero, leaving an equilibrium solution.

Conclusion

Comparing different Rossby numbers shows that rotation changes the shape of the motion in the rotating frame. For $R_0 = 10$ the orbit forms a star pattern since the direction of the back-and-forth motion drifts. For $R_0 = 2$ the trajectory becomes a diamond shape which is from more influence from rotation having bigger angular changes per swing. For $R_0 = 1$ with the given initial conditions and zero initial velocity, the motion stays at an equilibrium point and the pendulum does not swing.